

14-Watt Hi-Fi Audio Amplifier Using TDA2030A

Raj K. Gorkhali

Here is a simple mono audio amplifier using a single 5-pin IC TDA2030A. This IC is inexpensive and easily available in the market. The author's prototype, which gives 14W output, is shown in Fig. 1.

The circuit diagram of the audio amplifier is shown in Fig. 2. It comprises TDA2030A audio amplifier (IC1), two 1N4007 rectifier diodes (D1 and D2), four resistors (R1 through R4), nine capacitors (C1 through C9), 15-watt speaker (LS1), bridge rectifier (BR1) and a few other components.

For the power supply, the amplifier uses 230V AC primary to 12V-0-12V, 2A secondary transformer (X1). Two electrolytic capacitors C1 and C2 are used as filters for DC output voltage. The amplifier requires a dual $\pm 12V$ power supply, which can be obtained across transformer X1 and bridge rectifier BR1. The +12V

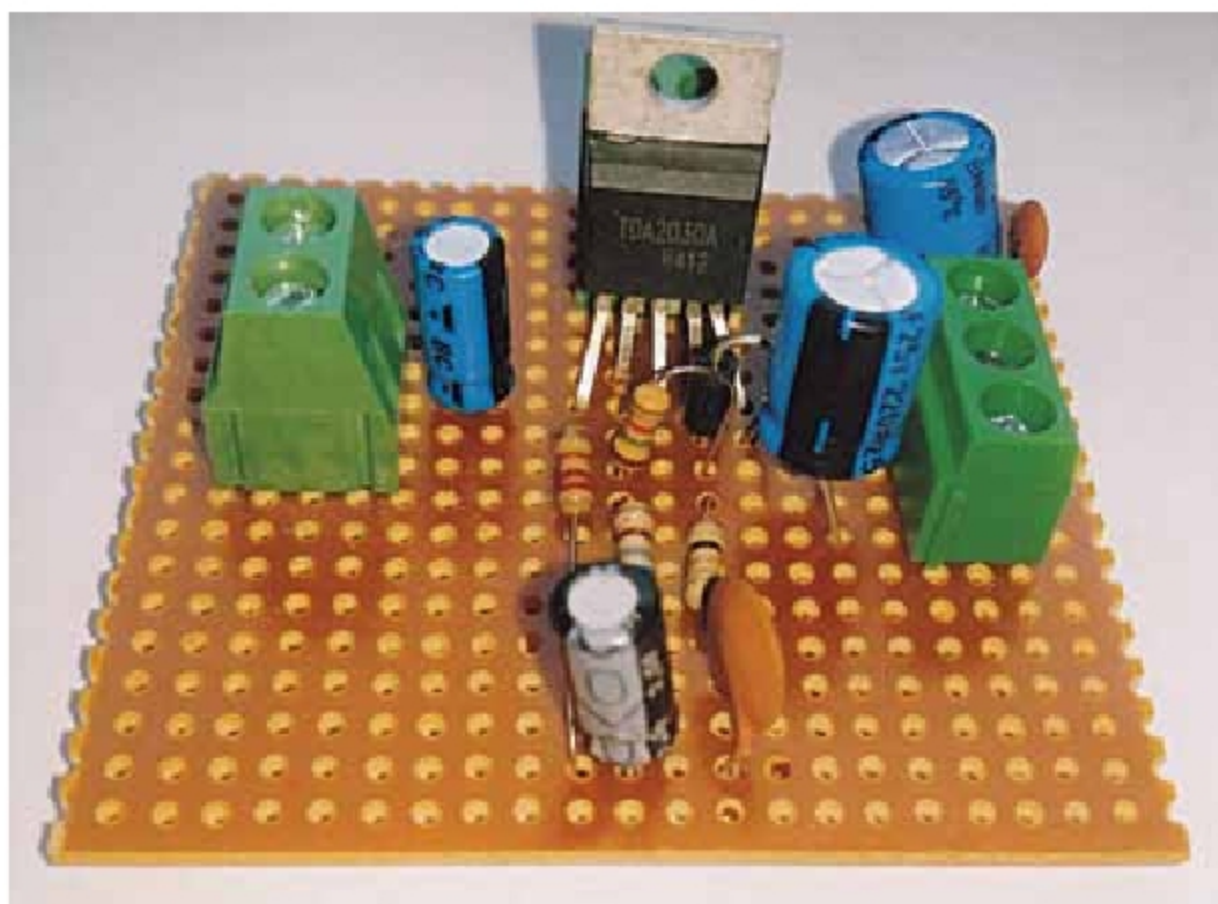


Fig. 1: Author's prototype

PARTS LIST

Semiconductors:

- IC1 - TDA2030A audio amplifier
- D1, D2 - 1N4007 rectifier diode
- BR1 - 2A bridge rectifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 22-kilo-ohm
- R2 - 680-ohm
- R3 - 12-kilo-ohm
- R4 - 1-ohm

Capacitors:

- C1, C2 - 1000 μ F, 35V electrolytic
- C3 - 1 μ F, 25V electrolytic
- C4, C7 - 100 μ F, 35V electrolytic
- C5, C8 - 100nF ceramic disk
- C6 - 22 μ F, 25V electrolytic
- C9 - 100nF ceramic disk

Miscellaneous:

- CON1 - 2-pin male Berg connector
- LS1 - 8-ohm, 15-watt speaker
- X1 - 230V AC primary to 12V-0-12V, 2A secondary transformer
- 2-pin terminal connector
- Heatsink for IC1

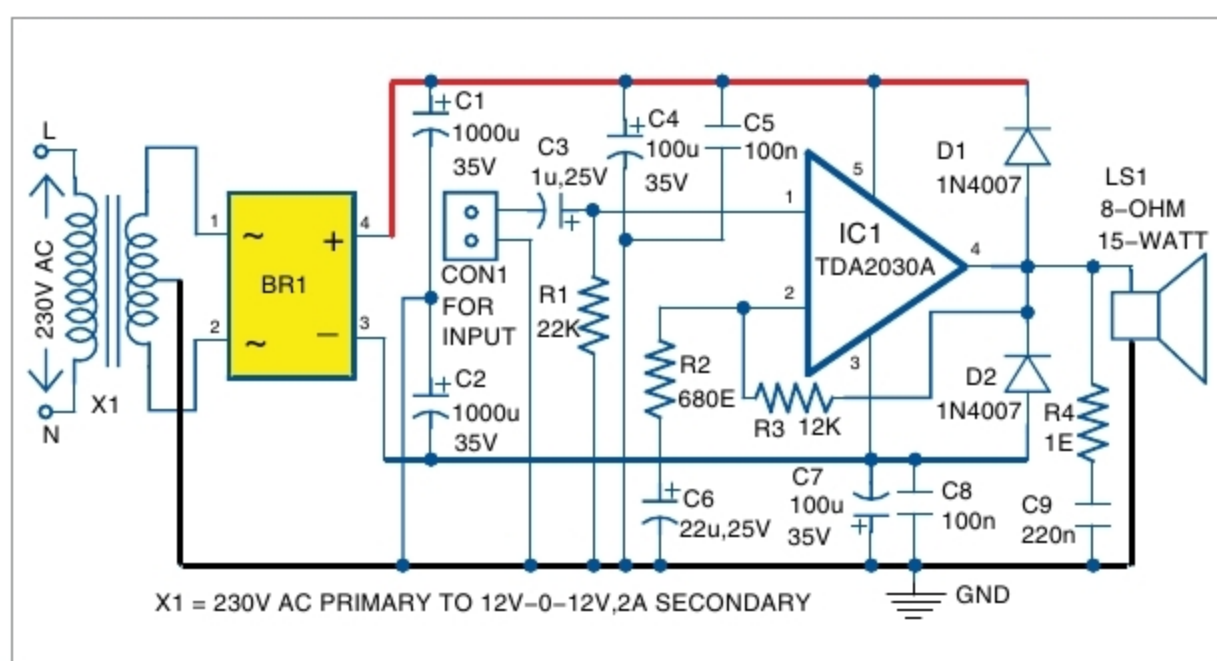


Fig. 2: Circuit diagram of the audio amplifier using TDA2030A

supply is connected to pin 5 and -12V to pin 3 of IC1.

Pin 1 of TDA 2030A is connected to the audio input terminal through electrolytic capacitor C3 (1 μ F/25V). Loudspeaker LS1 is connected to output pin 4 of IC1.

Construction and testing

An actual-size PCB layout for the audio amplifier circuit is shown in Fig. 3, and its components layout in Fig. 4. After assembling the circuit on PCB, connect audio input from computer/